

James Philip Iddon

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Valencia

Applied physicist specialising in the construction, verification, commissioning, and operation of large-scale silicon particle-tracking detectors and their data-acquisition systems, with experience across the ALICE, ATLAS, and Belle II collaborations at CERN and IFIC/Universitat de València. Research interests include silicon pixel and strip sensor characterisation and qualification, system-level verification and validation (V&V) of complex detector systems, and detector commissioning and operations.

Research Profile

- Silicon pixel and strip detector development and qualification
- Detector systems verification, validation, commissioning, and operations
- Data acquisition, hardware-software integration, and automated testing

Education

PhD in Applied Physics

University of Liverpool / CERN

Oct 2017 – Jul 2022

- Thesis: Construction, Commissioning and Performance Measurements of the Inner Tracking System Upgrade of ALICE at the LHC.
- Defended June 2022. CERN Doctoral Student Programme. Shortlisted for the ALICE thesis award.

MPHYS Physics

University of Liverpool

Sep 2013 – Jul 2017

- Integrated masters, specialised in detector physics • Grade: First Class
- Masters project: 'Inner Tracking System Upgrade of the ALICE Experiment at the LHC', characterisation of silicon CMOS MAPS chips.

Employment

Applied Physicist (Severo Ochoa Postdoctoral Fellow)

IFIC / Universitat de Valencia, Valencia, ES

Apr 2026 – Present

Applied Physicist (Staff)

CERN, Geneva, CH

Mar 2025 – Dec 2025

Applied Physicist (Senior Applied Fellow)

CERN, Geneva, CH

Jul 2022 – Feb 2025

Applied Physicist (Doctorant)

University of Liverpool / CERN

Oct 2017 – Jul 2022

Projects and Technical Contributions

Belle II VTX Upgrade — OBELIX CMOS Telescope (2026 – present)

IFIC / Universitat de Valencia

- Convener of the OBELIX CMOS telescope group in the Belle II VTX upgrade.

ATLAS ITk Strip Detector (2026 – present)

IFIC / Universitat de Valencia

- Construction and integration of the ITk Strip detector.
- ATLAS Physics Office Member.

ATLAS ITk Pixel Detector — Outer Barrel Verification and Validation (2025)

CERN

- Integration of the outer barrel of the ATLAS ITk Pixel detector, a system co-developed by 50 international institutions.

ATLAS Pixel Detector Operations (2022 – 2025)

CERN

- Mentored and led 24/7 shift teams drawn from a 300-person international pool.
- Chaired international coordination meetings across 9 time zones.

- Represented the ~30-person operations team within the 6000+ member ATLAS collaboration.
- Provided on-call operational and DAQ expertise for mission-critical operations over 2 years.
- Developed and maintained DAQ software (C++/Python, CI/CD pipelines) for continuous operations.

ALICE ITS2 — Inner Tracking System Upgrade (2017 – 2022)

University of Liverpool / CERN

- Constructed, characterised, and verified 25% of the Outer Barrel of the ALICE ITS2, the largest monolithic pixel sensor array ever built, with 98% manufacturing yield.
- Developer of verification software integrated with production readout and V&V strategies for the full detector system (12.5 billion pixels, 10m² silicon).
- Made the first measurement of detection efficiency using 5 million cosmic muon tracks across the full detector system.
- Presented technical work in international collaboration meetings (170+ institutions), from working groups to plenaries, and at national and international conferences.

Leadership

- Belle II VTX WG6 Convener, Jul 2026 - present
- ATLAS Pixel DAQ Coordinator, Oct 2024 - Jan 2025
- ATLAS Pixel Run Coordinator, Oct 2023 - Sep 2024
- ATLAS Pixel Deputy Run Coordinator, Apr 2023 - Sep 2023
- ALICE ITS2 Leader of the cosmic muon data collection campaign on surface, 2020

Awards and Honours

- 2025 Breakthrough Prize in Fundamental Physics, awarded jointly to the ALICE and ATLAS collaborations — one of 1,869 ALICE and 5,345 ATLAS researchers recognised.
- Shortlisted, ALICE Thesis Award, 2022.

Publications

All publications as an ALICE and ATLAS author can be found via my Inspire ID: [1618293](#), or via [ORCID](#).

Selected publications:

- ALICE ITS Collaboration [Sensor operating point calibration and monitoring of the ALICE Inner Tracking System during LHC Run 3](#), 2026
- The ATLAS collaboration [Sensor response and radiation damage effects for 3D pixels in the ATLAS IBL Detector](#), Journal of Instrumentation, 2024
- The ALICE collaboration, [ALICE upgrades during the LHC Long Shutdown 2](#), Journal of Instrumentation, 2024
- J. P. Iddon on behalf of the ATLAS collaboration, [Operational experience and performance with the ATLAS Pixel detector at the Large Hadron Collider at CERN](#), Nuclear Instruments and Methods in Physics Research Section A, 2024
- ALICE ITS Collaboration, [First demonstration of in-beam performance of bent Monolithic Active Pixel Sensors](#), Nuclear Instruments and Methods in Physics Research Section A, 2022
- J. P. Iddon on behalf of the ALICE collaboration, [Commissioning of the new ALICE Inner Tracking System](#), Journal of Instrumentation, 2020

Conference Presentations

- Operational Experience and Performance with the ATLAS Pixel detector at the Large Hadron Collider at CERN, HSTD13, Vancouver, 2023
- Realising the ALICE ITS Upgrade, IOP Nuclear Colloquium, Remote, Nov 2021
- Performance Measurements from Cosmic Muon Data using the Outer Barrel of the New ALICE Inner Tracking System, PSD12, Birmingham, Sept 2021
- Upgrade and Commissioning of the ITS Upgrade of ALICE, INSTR'20, Novosibirsk, Feb 2020
- Entering the precision era of hot QCD measurements, IOP18, Glasgow, Apr 2018

Conference / Workshop Organisation

- Organisation of Operations session, Pixel Week, Milan, 2024
- Organisation of Operations session, Pixel Week, Geneva, 2024

- Organisation of Operations session, Pixel Week, Paris, 2023

Supervision and Mentorship

- Two high school interns, Supervisor, High school internship, CERN, 2025
- One summer student, Co-supervisor, CERN Summer Student Program, CERN, 2025
- One PhD candidate, Technical co-supervisor, PhD Qualification Task, CERN, 2024
- One mentee, Mentor, Pixel Run Coordination, CERN, 2024
- Seven mentees, Mentor, Pixel Run Manager, CERN, 2023–2024
- One PhD candidate, Technical Supervisor, PhD Qualification Task, CERN, 2024
- One PhD candidate, Technical Supervisor, PhD Qualification Task, CERN (remote), 2023
- One Master's student, Assisted Supervision, Master's project, Daresbury Laboratory, 2019

Teaching

- ATLAS Run-3 Shift Training, Inner Detector Desk Training, 2024
- PHYS206 Practical Physics II, Teaching Assistant, 2017 - 2019

Professional Service

- Registered EU expert evaluator, European Commission
- Independent external expert for COST open calls

Affiliations

- IFIC, 2026 - present
- Belle II, 2026 - present
- Universitat de Valencia, 2026 - present
- ATLAS, 2022 - present
- CERN, 2016 - present
- ALICE ITS Collaboration, 2016 - 2023
- ALICE, 2016 - 2023
- University of Liverpool, 2011 - 2022
- Daresbury Laboratory, 2017 - 2019

Technical Skills

Technical: Python • C++ • Hardware-software integration • Verification & Validation (V&V) • CI/CD pipelines • Docker • Bash • Git • Linux • Automated testing and deployment

Systems Engineering: Systems lifecycle development, requirements analysis, integration, validation, deployment, and operational support • Configuration management and quality assurance

Communication & Coordination: Jira • International stakeholder engagement • Cross-functional team leadership • DevOps operations coordination

Code Quality: Code review workflows • Static analysis tools • Testing automation and continuous integration

Career Breaks

Intensive Spanish Course

Taronja Spanish School, Valencia

Dec 2025 – Apr 2026

- Full-time language study undertaken between the end of a CERN staff appointment and the start of a postdoctoral fellowship at IFIC/Universitat de Valencia.

Full driving licence • Spanish residence permit • English native speaker • Spanish A2+/B1